Databases
Further normalization

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Normalization

Requirement 1:
Your decomposition should be *lossless*

Requirement 2:
Your decomposition should avoid *redundancy*

Normalization: refreshing your memory

r		
A	**B**	**C**
a	10	100
b	20	200
a	10	300
b	20	400
c	30	500
b	20	600

We notice an FD: $A \rightarrow B$

Normalization: prevent redundancy caused by an FD

Requirement 2
Prevent redundancy caused by $A \rightarrow B$;
split r into r_1 and r_2

r		
A	**B**	**C**
a	10	100
b	20	200
a	10	300
b	20	400
c	30	500
b	20	600

r_1	
A	**B**
a	10
b	20
c	30

r_2	
A	**C**
a	100
b	200
a	300
b	400
c	500
b	600

Normalization: guarantee losslessness

Requirement 1

Guarantee losslessness by splitting over $A \rightarrow B$:

$$r_1 \bowtie r_2 = r$$

r_1

A	B
a	10
b	20
c	30

r_2

A	C
a	100
b	200
a	300
b	400
c	500
b	600

r

A	B	C
a	10	100
b	20	200
a	10	300
b	20	400
c	30	500
b	20	600

Normalization: the Decomposition Theorem

If we have a scheme $R(XYZ)$ and an FD $X \rightarrow Y$,

then the decomposition on $R_1(XY)$ and $R_2(XZ)$ is lossless

Avoiding redundancy: Boyce-Codd Normal Form

Suppose we have a relation schema R and a set FDs F

R is in <u>BCNF</u> (w.r.t. F) if each left side of a non-trivial FD is a superkey

Suppose we have a database schema DB, consisting of relation schemas $R_1, ..., R_n$

DB is in BCNF (w.r.t. F) if each relation schema R_i is in BCNF

Normalization

Algorithm:
INPUT: a universe R, a set FDs F
OUTPUT:
    a lossless BCNF-decomposition van R
METHOD:
    while there is a scheme S not in BCNF {
        suppose the villain has left side X;
        let $Y = X^+$ without X;
        let Z be the remaining attributes;
        split $S(XYZ)$ into $S_1(XY), S_2(XZ)$;
    }

Normalization

properties of BCNF

- BCNF represents the 'ultimate' level of redundancy prevention caused by FDs
- implementation: in a BCNF scheme, the left sides of FDs are keys, so you are able to enforce FDs by defining indices

What is an index?

```
SELECT * FROM Patient
   WHERE patientId = 3141592653
```

- An index on patientId supports direct access to the tuple satisfying the selection predicate
- Primary keys are automatically supported by an index
- Attributes often used in selections (e.g. birth_date, zipcode) might have an additional index
- Plural: indexes or indices

Observations

- the BCNF decomposition algorithm is not deterministic
- the number of possible BCNF decompositions may be very very large
- some BCNF decompositions turn out to be preferable to other BCNF decompositions

Back to the decomposition quality problem

Reminder: we have a relation scheme $R(ABCDE)$
and a set of fd's
$F = \{A \rightarrow BC, C \rightarrow D, D \rightarrow E\}$

Give at least two BCNF decompositions

Do you have a preference for one of the decompositions?

- (CD), (CE), (ABC)
- (DE), (CD), (ABC)

Let us try to to make the choice explicit

More observations

To support our search for *better* BCNF decompositions, we will need the notion of a

minimal representation of an FD set

FD sets may contain redundancy:

- $\{A \rightarrow B, A \rightarrow C, B \rightarrow C\}$
- $\{A \rightarrow B, AB \rightarrow C\}$
- $\{A \rightarrow BC, B \rightarrow C\}$

Types of redundancy in FD sets

- redundant FD:
  $\{A \rightarrow B, A \rightarrow C, B \rightarrow C\}$
  $A \rightarrow C$ is redundant
- reducable left side:
  $\{A \rightarrow B, AB \rightarrow C\}$
  B is superfluous in $AB \rightarrow C$
- reducable right side:
  $\{A \rightarrow BC, B \rightarrow C\}$
  C is superfluous in $A \rightarrow BC$

Maximal representation of FD set: closure

Definition:

F^+ is the set of all FD's derivable from F

We call F^+ the <u>closure</u> of F

don't try to calculate F^+ at home

Definition:

Two FD-sets F, G are <u>equivalent</u> if $F^+ = G^+$

INTERMEZZO

Suppose $F = \{A_1 \rightarrow A_2, A_2 \rightarrow A_3, \ldots, A_{n-1} \rightarrow A_n, A_n \rightarrow A_1\}$

Give an estimation (or a lower limit) for $|F^+|$ (the size of F^+).

What about $n = 100$?

Concise representation of FD set

Definition:

G is a <u>minimal cover</u> of F
if F and G are equivalent and

1. G does not contain redundant FDs
2. all right sides in G are minimal
3. all left sides in G are minimal

Terminology: minimal basis = minimal cover

Minimal cover calculation

Algorithm

INPUT: a set FDs F
OUTPUT: a minimal cover for F
METHOD:

1. split each FD into single right side FDs
2. reduce left sides
3. eliminate redundant FDs
4. (not mandatory) combine FDs with identical left sides

Minimal cover calculation

2. *reduce left sides:*

```
for each FD U → V in F {
    for each attribute A in U {
        let U' = U − {A};
        if U' → V can be derived from F
        then replace U → V with U' → V;
    }
}
```

Minimal cover calculation

3. *eliminate redundant FDs:*

for each FD $U \rightarrow V$ in F {
    let $F' = F - \{U \rightarrow V\}$;
    if $U \rightarrow V$ can be derived from F'
    then delete $U \rightarrow V$ from F
};

INTERMEZZO

We have a relation scheme R(ABCDEFGHKL)
and a set FD's
$\mathbf{F} = \{A \rightarrow D, B \rightarrow HA, GH \rightarrow AKL, CH \rightarrow BK, G \rightarrow BC\}$

Calculate a minimal cover for **F**

Step 1: split into single right side FD's

$A \rightarrow D$
$B \rightarrow H$
$B \rightarrow A$
$GH \rightarrow A$
$GH \rightarrow K$
$GH \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 2: reduce left sides

Can $GH \rightarrow A$ be reduced to $G \rightarrow A$?

$A \rightarrow D$
$B \rightarrow H$
$B \rightarrow A$
$GH \rightarrow A$
$GH \rightarrow K$
$GH \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 2: reduce left sides

Can $GH \rightarrow A$ be reduced to $G \rightarrow A$?
$G^+ = GBCHAKLD$, so $G \rightarrow A$ holds!

$A \rightarrow D$
$B \rightarrow H$
$B \rightarrow A$
$G \rightarrow A$
$GH \rightarrow K$
$GH \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 2: reduce left sides

$G^+ = GBCHAKLD$, so $G \rightarrow KL$ also holds

$A \rightarrow D$
$B \rightarrow H$
$B \rightarrow A$
$G \rightarrow A$
$G \rightarrow K$
$G \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 2: reduce left sides

Can $CH \rightarrow B$ be reduced to $C \rightarrow B$?
$C^+ = C$, so $C \rightarrow B$ does not hold!
Neither do $H \rightarrow B, C \rightarrow K, H \rightarrow K$

$A \rightarrow D$
$B \rightarrow H$
$B \rightarrow A$
$G \rightarrow A$
$G \rightarrow K$
$G \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 3: eliminate redundant FD's

Can $A \rightarrow D$ be eliminated?

$A \rightarrow D$
$B \rightarrow H$
$B \rightarrow A$
$G \rightarrow A$
$G \rightarrow K$
$G \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 3: eliminate redundant FD's

Can $A \rightarrow D$ be eliminated?
Try to derive $A \rightarrow D$ from the remaining FD set
$\mathbf{F}' = \mathbf{F} - \{A \rightarrow D\}$.
Within $\mathbf{F}'$, $A^+ = A$, so $A \rightarrow D$ cannot be derived.

$A \rightarrow D$

$B \rightarrow H$
$B \rightarrow A$
$G \rightarrow A$
$G \rightarrow K$
$G \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 3: eliminate redundant FD's

Can $B \rightarrow H$ be eliminated?

$A \rightarrow D$

$B \rightarrow H$

$B \rightarrow A$
$G \rightarrow A$
$G \rightarrow K$
$G \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 3: eliminate redundant FD's

Can $B \rightarrow H$ be eliminated?
Try to derive $B \rightarrow H$ from $\mathbf{F}' = \mathbf{F} - \{B \rightarrow H\}$.
Within $\mathbf{F}'$, $B^+ = BAD$, so $B \rightarrow H$ cannot be derived.

$A \rightarrow D$

$B \rightarrow H$

$B \rightarrow A$
$G \rightarrow A$
$G \rightarrow K$
$G \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 3: eliminate redundant FD's

Can $G \rightarrow A$ be eliminated?

$A \rightarrow D$
$B \rightarrow H$
$B \rightarrow A$

$G \rightarrow A$

$G \rightarrow K$
$G \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 3: eliminate redundant FD's

Can $G \rightarrow A$ be eliminated?
Try to derive $G \rightarrow A$ from $\mathbf{F}' = \mathbf{F} - \{G \rightarrow A\}$.
Within $\mathbf{F}'$, $G^+ = GKLBCDHA$, so $G \rightarrow A$ can be derived.

$A \rightarrow D$
$B \rightarrow H$
$B \rightarrow A$

$G \rightarrow A$

$G \rightarrow K$
$G \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

Step 3: eliminate redundant FD's

Final result, minimal cover

$A \rightarrow D$
$B \rightarrow H$
$B \rightarrow A$
$G \rightarrow L$
$CH \rightarrow B$
$CH \rightarrow K$
$G \rightarrow B$
$G \rightarrow C$

---

or, equivalently: $\{A \rightarrow D, B \rightarrow HA, G \rightarrow LBC, CH \rightarrow BK\}$

Back to the decomposition quality problem

Reminder: we have a relation scheme $R(ABCDE)$
and a set of fd's
$F = \{A \rightarrow BC, C \rightarrow D, D \rightarrow E\}$

Give at least two BCNF decompositions

Do you have a preference for one of the decompositions?

- (CD), (CE), (ABC)
- (DE), (CD), (ABC)

The second solution is *dependency preserving*

Projection of FD's

Definition:

The projection of FD $U \rightarrow V$ on scheme R is:

1. $U \rightarrow V$, if $UV \subseteq attr(R)$
2. void, if one of the attributes (left or right) is not in $attr(R)$

Dependency preserving decompositions

Definition:
Suppose we have a scheme R and a set FDs F.
A decomposition of R into R_1, R_2 is called
<u>dependency preserving</u> (DP) if:

$$(F_1 \cup F_2)^+ = F^+$$

where F_i is the projection of F^+ on R_i

Dependency preserving decomposition: examples

Suppose we have a scheme $R(ABCDE)$, with
$F = \{A \rightarrow BE, C \rightarrow DE\}$

Decomposition step 1:
$R_1(ABC), R_3(CDE)$

Decomposition step 2:
$R_1(AB), R_2(AC), R_3(CDE)$

$F_1 = \{A \rightarrow B, ...\}, F_2 = \{...\}, F_3 = \{C \rightarrow D, C \rightarrow E, ...\}$

Where is $A \rightarrow E$?

Can it be derived?

Dependency preserving decomposition: examples

Suppose we have a scheme $R(ABCDE)$, with
$F = \{A \rightarrow BE, C \rightarrow DE\}$

Decomposition step 1:
$R_1(ABC), R_3(CDE)$

Decomposition step 2:
$R_1(AB), R_2(AC), R_3(CDE)$

$F_1 = \{A \rightarrow B, ...\}, F_2 = \{...\}, F_3 = \{C \rightarrow D, C \rightarrow E, ...\}$

Where is $A \rightarrow E$?

Can it be derived? No!

This decomposition is not DP!

Dependency preserving decomposition: examples

Suppose we have a scheme $R(ABCDE)$, with
$F = \{A \rightarrow BCE, C \rightarrow DE\}$

Decomposition step :
$R_1(ABC), R_2(CDE)$

$F_1 = \{A \rightarrow B, A \rightarrow C, ...\}, F_2 = \{C \rightarrow D, C \rightarrow E, ...\}$

Where is $A \rightarrow E$?

Dependency preserving decomposition: examples

Suppose we have a scheme $R(ABCDE)$, with
$F = \{A \rightarrow BCE, C \rightarrow DE\}$

Decomposition step :
$R_1(ABC), R_2(CDE)$

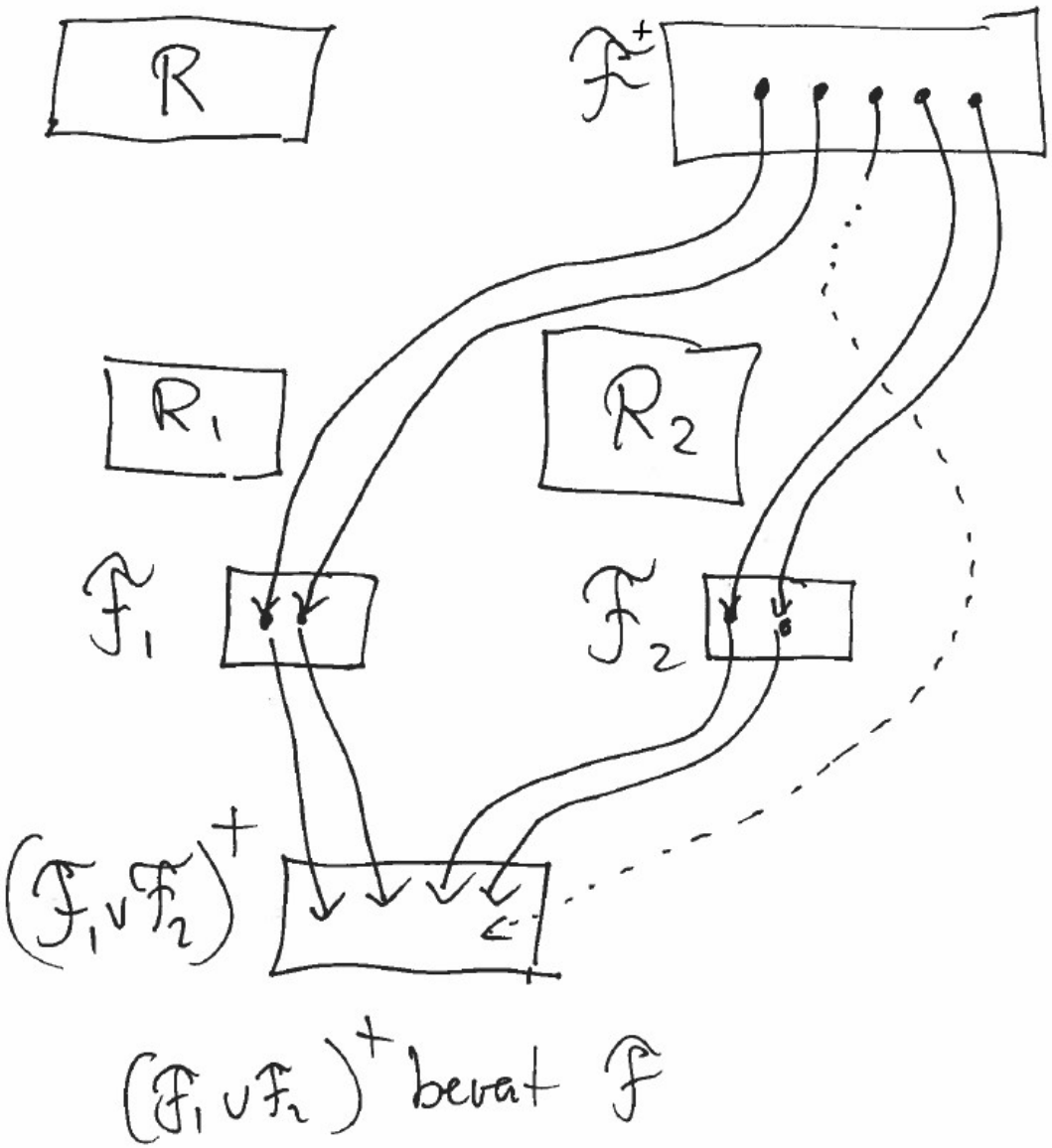
$F_1 = \{A \rightarrow B, A \rightarrow C, ...\}, F_2 = \{C \rightarrow D, C \rightarrow E, ...\}$

Where is $A \rightarrow E$?

It can be found in $(F_1 \cup F_2)^+$!

This decomposition is DP!

DPdecomposition: artist's impression



INTERMEZZO

$R = (ABCDE)$
$F = \{A \rightarrow BC, D \rightarrow E\}$
$R_1 = (ABC), R_2 = (ADE)$
Is this decomposition lossless, BCNF, DP?

$R = (ABCDE)$
$F = \{A \rightarrow BC, D \rightarrow E\}$
$R_1 = (AB), R_2 = (AC), R_3 = (ADE)$
Is this decomposition lossless, BCNF, DP?

$R = (ABCDE)$
$F = \{A \rightarrow BCDE, C \rightarrow A, D \rightarrow E\}$
$R_1 = (ABC), R_2 = (CD), R_3 = (DE)$
Is this decomposition lossless, BCNF, DP?

Observations

- a DP decomposition is preferable to a non DP decomposition, because it enables efficient FD checking
- a DP/BCNF decomposition does not always exist
- the BCNF decomposition algorithm does not produce always DP decompositions (even if they do exist)

Normalization

Making a choice

- Minimize the level of redundancy: BCNF
- Accept that the DP property may be lost

or:

- Allow a bit more redundancy (3NF in stead of BCNF)
- Enforce the DP property

3NF (just out of the blue)

Definition:
An attribute is *prime* if it is contained in a candidate key.

Definition:
A relation scheme is in <u>3NF</u> if for each non trivial FD $X \rightarrow A$ the following condition holds:

$$X \text{ is a superkey } \textit{or } A \text{ is prime}$$

3NF: another view

A relation scheme is in 3NF if

- it is in BCNF, or
- if all FDs that violate the BCNF property have this shape: $X \rightarrow A$ where X is not a key, but A is part of a key

Note that the 3NF requirement is a bit less demanding than the BCNF requirement

BCNF versus 3NF

Address table for a specific town:

ADDRESS		
Street	**Number**	**Zipcode**
Ooievaarspad	1	3403 AM
Ooievaarspad	2	3403 AM
Meerkoetweide	2	3403 AK
Meerkoetweide	4	3403 AK
Meerkoetweide	6	3403 AL
Meerkoetweide	8	3403 AL

BCNF versus 3NF

ADDRESS		
Street	**Number**	**Zipcode**
Ooievaarspad	1	3403 AM
Ooievaarspad	2	3403 AM
Meerkoetweide	2	3403 AK
Meerkoetweide	4	3403 AK
Meerkoetweide	6	3403 AL
Meerkoetweide	8	3403 AL

Street, Number $\rightarrow$ Zipcode
Zipcode $\rightarrow$ Street
ADDRESS is in 3NF
ADDRESS is not in BCNF (redundancy)

BCNF versus 3NF

StrZip	
Street	**Zipcode**
O'pad	3403 AM
M'weide	3403 AK
M'weide	3403 AL

NoZip	
Number	**Zipcode**
1	3403 AM
2	3403 AM
2	3403 AK
4	3403 AK
6	3403 AL
8	3403 AL

StrZip and NoZip are in BCNF
This decomposition is not DP
An operation `insert(6,'3403 AK')` on NoZip is incorrect!
Detecting this violation requires a join with StrZip

3NF-Algorithm

INPUT: a scheme R, a set FDs F
OUTPUT: a lossless DP 3NF-decomposition of R
METHOD:

create a minimal cover G from F;
generate for each FD $X \rightarrow A_1, ..., A_n$ a scheme $(XA_1A_2..A_n)$;
// this scheme has local key X
IF there is a scheme containing a global key K
THEN you are finished;
ELSE add a global key as an extra relation scheme;
// a global key is a key for R

INTERMEZZO

- We have $R(ABCDEFGHKL)$ with
  $\mathbf{F} = \{A \rightarrow D, B \rightarrow HA, GH \rightarrow AKL, CH \rightarrow BK, G \rightarrow BC\}$
- Minimal cover
  $\mathbf{G} = \{A \rightarrow D, B \rightarrow HA, G \rightarrow LBC, CH \rightarrow BK\}$
- 3NF scheme?

INTERMEZZO

- We have $R(ABCDEFGHKL)$ with
  $\mathbf{F} = \{A \rightarrow D, B \rightarrow HA, GH \rightarrow AKL, CH \rightarrow BK, G \rightarrow BC\}$
- Minimal cover
  $\mathbf{G} = \{A \rightarrow D, B \rightarrow HA, G \rightarrow LBC, CH \rightarrow BK\}$
- 3NF scheme:
  $(AD), (BHA), (GLBC), (CHBK)$

INTERMEZZO

- We have $R(ABCDEFGHKL)$ with
  $\mathbf{F} = \{A \rightarrow D, B \rightarrow HA, GH \rightarrow AKL, CH \rightarrow BK, G \rightarrow BC\}$
- Minimal cover $\mathbf{G} = \{A \rightarrow D, B \rightarrow HA, G \rightarrow BCL, CH \rightarrow BK\}$
- 3NF scheme:
  $(AD), (BHA), (GBCL), (CHBK)$
- Lossless 3NF scheme:
  $(AD), (BHA), (GBCL), (CHBK), (GEF)$

Overview normal forms

normal form	*feasible*	*complexity*
3NF + DP	always	polynomial
BCNF	always	polynomial
BCNF + DP	not always	NP-hard

Requirement 2:
you should strive for a high normal form (BCNF or 3NF)
Requirement 3:
you should strive for a DP decomposition